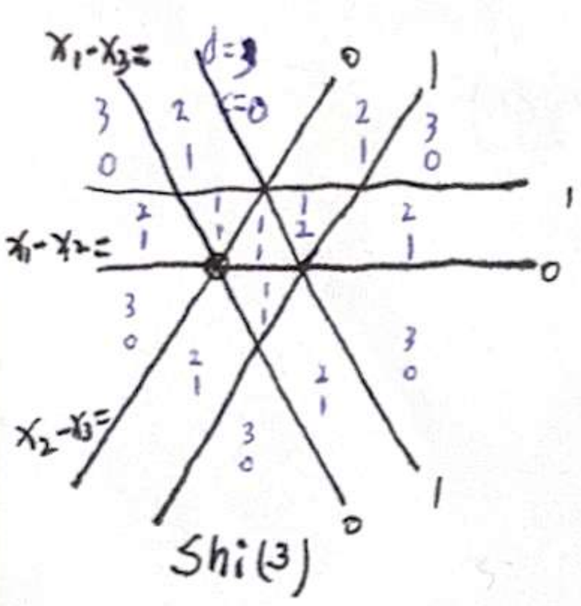
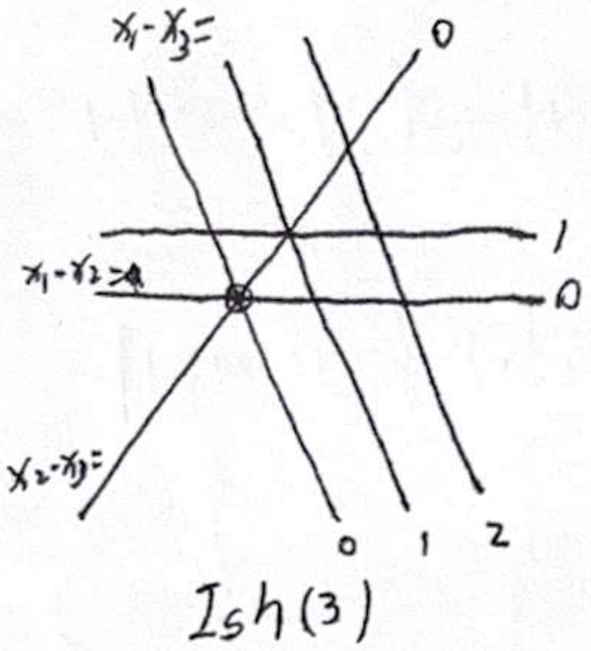
Coxeter arrangement of type A:
余元斯答
$$Cox(n) := \{x_i - x_j = 0 : 1 \le i < j \le n\}$$

Shi arrangement:
$$Shi(n) := Cox(n) \cup \{x_i - x_j = 1 : 1 \le i < j \le n\}$$

Ish arrangement:
$$Ish(n) := Cox(n) \cup \{x_i - x_j = i : 1 \le i < j \le n\}$$



Shi(3)



Ish(3)

combinatorial symmetric:

$x_i - x_j = 1 \iff x_1 - x_j = i$ for $1 \le i < j \le n$

c \ d	1	2	3
0			6
1	3	6	
2	1		

The deleted version:
$$Shi(G) := Cox(n) \cup \{x_i - x_j = 1 : ij \in G\}$$
$$Ish(G) := Cox(n) \cup \{x_i - x_j = i : ij \in G\}$$

Regions: The connected components of $\mathbb{R}^n - \bigcup_{H \in \mathcal{A}} H$.

A region is dominant if it lies with the dominant cone. $x_1 > x_2 > \cdots > x_n$

Faces: The topological closure $\bar{R}$ of a region R is decomposed by the arrangement $\mathcal{A}$ into faces of verious dimensions.

Wall: If it's the affine span of a codimension-1 faces of R

Ceiling: A wall ① doesn't contain the origin.
② the region R and the origin lies in the same half-space.

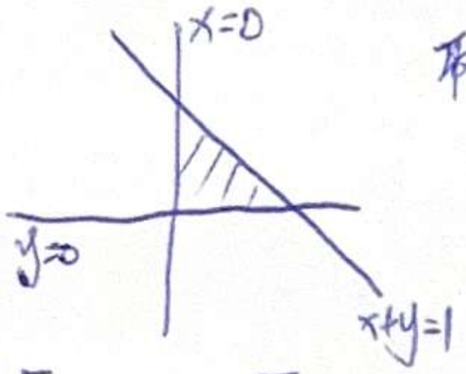
Recession cone: (is closed under nonnegative linear combonations)
$$Rec(R) := \{v \in \mathbb{R}^n : v + R \subseteq R\}$$
R is bounded iff $Rec(R) = 0$
$\dim(Rec(R)) =$ the number of degrees of freedom of R

$L(\mathcal{A}) := \{\bigcap_{H \in S} H : S \subseteq \mathcal{A}\}$ is partially ordered by the reverse-inclusion of ~~superspace~~ subspace.
$X \le Y \Leftrightarrow X \supseteq Y$

Characteristic polynomial: $\chi_{\mathcal{A}}(p) := \sum_{X \in L(\mathcal{A})} \mu(\mathbb{R}^n, X)\, p^{\dim X} \in \mathbb{Z}[p]$

$\mu : L(\mathcal{A}) \times L(\mathcal{A}) \to \mathbb{Z}$ is the Möbius function of poset $L(\mathcal{A})$
$\mu(x,x) = 1$
$x < y$ 时, $\sum_{x \le z \le y} \mu(x,z) = 0 \iff \mu(x,y) = -\sum_{x \le z < y} \mu(x,z)$

e.g. $\mathbb{R}^2$ $\mathcal{A}=\{H_1, H_2, H_3\}$ $H_1: x=0$ $H_2: y=0$ $H_3: x+y=1$



取 $R=\{(x,y): x>0, y>0, x+y<1\}$

Face: $\overline{R}=\{(x,y): x\geq 0, y\geq 0, x+y\leq 1\}$

dim=2: $\{x>0, y>0, x+y<1\}$

dim=1: $F_1=\{(0,y): 0<y<1\}$ $F_2=\{(x,0): 0<x<1\}$ $F_3=\{(x,y): x+y=1, 0<x<1, 0<y<1\}$

dim=0: $(0,0)$ $(0,1)$ $(1,0)$

Wall: 一维face F_3, affine span: $\mathrm{aff}(F_3)=\{(x,y): x+y=1\}$

$y=0$, $x=0$

ceiling: $x+y=1$

recession cone: $\mathrm{Rec}(R)=\emptyset$

取 $R'=\{(x,y): x>0, y>0, x+y>1\}$ $\mathrm{Rec}(R')=\{(a,b): a\geq 0, b\geq 0\}$

$\dim(R')=2$ = degree of freedom of R'

e.g. $\mathcal{A}=\{H_1, H_2\}$ $H_1=\{x=0\}$ $H_2=\{y=0\}$

$L(\mathcal{A})=\{\mathbb{R}^2, H_1, H_2, \{(0,0)\}\}$ $\mathbb{R}^2 < H_1, H_2 < \{(0,0)\}$

$\mu(\mathbb{R}^2,\mathbb{R}^2)=1$ $\sum_{\mathbb{R}^2\leq Z\leq H_1} \mu(\mathbb{R}^2, Z)=\mu(\mathbb{R}^2,\mathbb{R}^2)+\mu(\mathbb{R}^2,H_1)=0 \Rightarrow \mu(\mathbb{R}^2,H_1)=-1$

$\Rightarrow \mu(\mathbb{R}^2,H_2)=-1$

$1+(-1)+(-1)+\mu(\mathbb{R}^2,\{(0,0)\})=0 \Rightarrow \mu(\mathbb{R}^2,\{(0,0)\})=1$

$\Rightarrow \chi_{\mathcal{A}}(p)=p^2-p-p+1=(p-1)^2$

$$\#\{\text{regions of } \mathcal{A}\}=(-1)^n \chi_{\mathcal{A}}(-1)$$

Main Theorem:

$G\subseteq\binom{[n]}{2}$ be a simple loopless graph on n vertices, $c,d\in\mathbb{Z}_{\geq 0}$.

The deleted Shi and Ish arrangements $\mathrm{Shi}(G)$ and $\mathrm{Ish}(G)$ share the following properties in common.

(1) the characteristic polynomial;

(2) the number of dominant regions with c ceiling;

(3) the number of regions with c ceilings and d degrees of freedom.

Set partitions

$\pi = \{B_1, \cdots, B_k\}$ is a partition of $[n]$ into k blocks

$[n] = B_1 \sqcup B_2 \sqcup \cdots \sqcup B_k$ (disjoint union).

type: sequence $(r_1, \cdots, r_n)$ r_i is the number of blocks of π with size i.

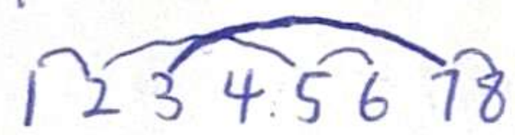
arc diagram of π: ① place the number $1, 2, \cdots, n$ on a line

② draw an arc between each pair $i<j$ such that:

1) i, j in the same block of π;

2) no $i<\ell<j$ s.t. i, ℓ, j are in the same block.

eg. $\{\{1,2,5,6\}, \{3,7,8\}, \{4\}\}$ type $(1,0,1,1,0,0,0,0)$



$a = (1\,2\,3\,5\,7)$ $b = (2\,5\,6\,8)$

π has $n-k$ blocks iff its diagram has k arcs

Suppose that arcs of π are $a_1b_1, a_2b_2, \cdots, a_kb_k$, $a_1 < a_2 < \cdots < a_k$

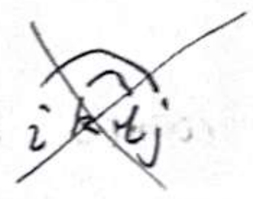
$\pi \to (a,b)$: $a = a_1 a_2 \cdots a_k$ and $b = b_1 b_2 \cdots b_k$

(end point rotation)

partition of $[n]$ are bijection with (a,b) s.t.:

- a, b have the same length
- $a_i < b_i$ $\forall i$
- the entries of a are increasing
- the entries of b are distinct

Nonnesting partitions:



π of $[n]$ is called nonnesting if it doesn't contain arcs ij and kl s.t. $i<k<l<j$

The number of nonnesting partitions of $[n]$ is famously given by Catalan number $\frac{1}{n+1}\binom{2n}{n}$

(a,b) is nonnesting iff its right endpoint vector b is increasing.

the number of pairs of nesting arcs in (a,b) is equal to the number of pairs $b_i > b_j$ s.t. $i<j$.

G-partitions and G-stirling numbers:

π of $[n]$ is ~~called~~ a G-partition if all of its arcs are contained in G.

G-stirling number $Stir(G,k)$ is the number of G-partition with k-blocks.

Connectivity

For $i<j$, there doesn't exist $i \le k < j$ s.t. $\{\{i, i+1, \cdots, k\}, \{k+1, \cdots, j-1, j\}\}$, called $\{i, \cdots, j\}$ is connectivity.

i.e. arc diagram has no holes.

If exist $1 < i_1 < \cdots < i_{d-1} < n$ s.t. π refines the partition

$\{\{1, 2, \cdots, i_1-1\}, \{i_1, \cdots, i_2-1\}, \cdots, \{i_{d-1}, \cdots, n\}\}$ each blocks is connected,

π has d connected components. $\longrightarrow$ arc diagram has $d-1$ holes.

3. Characteristic polynomial. (finite field method).

The Method:

Let $p \in \mathbb{Z}$ be prime and consider a hyperplane $H \subseteq \mathbb{R}^n$ with fixed equation

$a_1 x_1 + \cdots + a_n x_n = b$. where $a_i, b \in \mathbb{Z}$

※ Define H_p of the finite vector space $\mathbb{F}_p^n$ by reducing the coefficients of H module p:

$$H_p := \{(x_1, x_2, \cdots, x_n) \in \mathbb{F}_p^n : \bar{a}_1 x_1 + \cdots + \bar{a}_n x_n = \bar{b}\}$$

$a_i \to \bar{a}_i \in \mathbb{F}_p$

$b \to \bar{b} \in \mathbb{F}_p$.

If p is small $\longrightarrow$ H_p may not hyperplane

$a_i \to 0$

$H = \{x \in \mathbb{R} : 2x = 1\}$

$p = 2$ $0 = 1$ X.

We want p is large enough.

$\mathcal{A}_p := \{H_p : H \in \mathcal{A}\}$ is called **the reduced hyperplane arrangement**

Thm 3.1 (Crapo and Rota)

Let $p \in \mathbb{Z}$ be a large prime, and let $\mathcal{A}$ be a finite collection of hyperplanes in $\mathbb{R}^n$ whose hyperplanes have defining equations with coefficients in $\mathbb{Z}$.

Then the characteristic polynomial of $\mathcal{A}$ satisfies

$$\chi_{\mathcal{A}}(p) = \#\left(\mathbb{F}_p^n - \bigcup_{H \in \mathcal{A}} H_p\right)$$

$p \in \mathbb{Z}$为充分大素数，$\mathcal{A}$为$\mathbb{R}^n$中有限超平面的集合
其定义方程系数为整数

$\chi_{\mathcal{A}}(p)$ counts the number of points in the complement of the reduced arrangement $\mathcal{A}$. 补集的点数

Thm 3.2

Let $G \subseteq \binom{[n]}{2}$ be a graph on n vertices. The characteristic polynomial of the deleted Shi and Ish arrangement are given by

$$\chi_{Shi(G)}(p) = \chi_{Ish(G)}(p) = p\sum_{k=0}^{n-1}(-1)^k Stir(G, n-k)\frac{(p-k-1)!}{(p-n)!}$$

Pf: We need to proof: $\#(\mathbb{F}_p^n - Shi(G)_p) = \#(\mathbb{F}_p^n - Ish(G)_p)$

Let $\{0,1,\cdots,n-1\} = \mathbb{F}_p$ as the vertices of regular p-gon. (ordered clockwise) $i \to i+1$

And $v=(v_1,\cdots,v_n)\in\mathbb{F}_p^n$ is a labeling of vertices: If $v_i=j$, place the label i on vertex j.

$v \in \mathbb{F}_p^n - Cox(n)_p \Leftrightarrow v_i - v_j \neq 0$ for all $1\le i<j\le n$. $\to i\neq j \Rightarrow v_i \neq v_j$

i.e. the points of $\mathbb{F}_p^n - Cox(n)_p$ corrospond to injective labelings $\{v_1,\cdots,v_n\} \to \mathbb{F}_p$.

$Shi(G)_p$ and $Ish(G)_p$ are both contained in $\mathbb{F}_p^n - Cox(n)_p$.

First for $Shi(G)_p$, 表示对所有边 $ij\in S$ 都有 $v_i - v_j = 1$ 的向量 $v\in\mathbb{F}_p^n - Shi(G)_p$ 的数量.

Let $f(S)$ denote the number of vertors $v\in\mathbb{F}_p^n - Shi(G)_p$ s.t. $v_i - v_j = 1$ for all $ij\in S$ $(i<j)$

$U = \mathbb{F}_p^n - Cox(n)_p$, $A_{ij} = \{v\in U : v_i - v_j = 1\}$

$\mathbb{F}_p^n - Shi(G)_p = U - \bigcup_{ij\in G} A_{ij}$ $\quad \left|U - \bigcup_{ij\in G} A_{ij}\right| = \sum_{S\subseteq G}(-1)^{|S|}\left|\bigcap_{ij\in S} A_{ij}\right|$ $\to f(S)$

\# By the principle of inclusion-exclusion

So the number of points in $\mathbb{F}_p^n - Shi(G)_p$ is equal to $\sum_{S\subseteq G}(-1)^{|S|}f(S)$.

Focus on S: If S contain edges ij and il with the same left point endpoint, we have

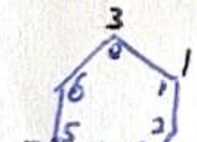
$\begin{cases} v_i - v_j = 1 \\ v_i - v_l = 1 \end{cases} \Rightarrow v_j = v_l$ $\times$ $\Rightarrow f(S) = 0$ (right.)

$\Rightarrow$ The sets S that contribute to the sum is precisely the arc sets of G-partitions.

Then compute $f(S)$ For fixed number of blocks $(n-k)$.

e.g. $n=5$ $\pi = \{\{13\},\{24\},\{5\}\}$ $n-k=3$ blocks $k=2$ arcs $\Rightarrow$ arc set: $S=\{13, 24\}$

$v_1 - v_3 = 1$, $v_2 - v_4 = 1$

$p=7$ 

$v_1 = v_3 + 1$ 1放的位置是3的下一个

(31) (42) (5) strings of labels

$v_1 = 1$, $v_2 = 4$, $v_3 = 0$, $v_4 = 3$, $v_5 = 5$ $\Rightarrow v = (1\,4\,0\,3\,5)$



p points

$p-n$ empty points

$n-k$ strings of label $\Rightarrow \binom{(p-n)+(n-k)-1}{(n-k)-1} = \binom{p-k-1}{n-k-1} = \frac{(p-k-1)!}{(n-k-1)!\,(p-n)!}$.

\# $n-k$ 个间隔

There are $(n-k-1)!$ ways to cyclically permute the strings. $e_1 + e_2 + \cdots + e_{n-k} = p-n$

There are p ways to choose the location of 0. Hence $f(S) = \frac{p(p-k-1)!}{(p-n)!}$ 放的位置数 factorial.

$$\chi_{Shi(G)}(p) = \sum_{S\subseteq G}(-1)^{|S|}f(S)$$

$$= \sum_{k=0}^{n-1}(-1)^k Stir(G,n-k)\frac{p\cdot(p-k-1)!}{(p-1)!}$$

$Stir(G,n-k)$ → the number of G-partition with $n-k$ blocks.

For $Ish(G/p)$, is similar.

Let $g(S)$ be the number of vectors $v \in \mathbb{F}_p^n - Ish(G/p)$ s.t. $v_i - v_j = i$, $\forall ij \in S$.

Then the points of $\mathbb{F}_p - Ish(G/p)$: $\sum_{S \subseteq G} (-1)^{|S|} g(S)$

$\left.\begin{matrix} v_i - v_j = i \\ v_i - v_k = i \end{matrix}\right\} \Rightarrow v_j = v_k \Rightarrow g(S) = 0$ unless S is the arc set of a G-partition.

First choose v_1 in p ways. The left $(n-k-1)$ labels must be placed injectively in the remaining $p-k-1$ positions.

Also for fixed $(n-k)$ blocks.

There are $(p-k-1)(p-k-2)\cdots(p-n+1)$ ways to do.

we get $g(S) = \frac{p(p-k-1)!}{(p-n)!}$. ■

When $G = K_n$, $Stir(G, n-k) = S(n, n-k)$ the second Stirling number

$$\chi(p) = p \sum_{k=0}^{n-1} (-1)^k S(n, n-k) \frac{(p-k-1)!}{(p-n)!}$$

Let $r = n-k$ $x = p-n$ $\Rightarrow \frac{(p-k-1)!}{(p-n)!} = \frac{x^{\bar{r}}}{x}$ → rising factorial

$x^{\bar{r}} := x(x+1)\cdots(x+r-1)$

$$\chi(p) = \frac{p}{x} \sum_{r=1}^{n} (-1)^{n-r} S(n, r) x^{\bar{r}}$$

Use identity: $\boxed{\sum_{r=0}^{n} (-1)^{n-r} S(n, r) x^{\bar{r}} = x^n}$ $\Rightarrow \chi(p) = \frac{p}{x} x^n = p x^{n-1} = p(p-n)^{n-1}$

Characteristic polynomial of an arrangement allows us to count certain kinds of regions.

Some notation:

Let $\mathcal{A}$ be a finite collection ~~of count certain~~ of hyperplane in $\mathbb{R}^n$

the normals to the hyperplanes span a space $V \subseteq \mathbb{R}^n$ of dimension r.

r is called the rank of the arrangement $r \leq n$

If $r < n$, the $\mathcal{A}$ has no bounded regions.

If a region's intersection with V is bounded, we say the region is relatively bounded.

Thm 3.4 (Zaslavsky)

Let $\mathcal{A}$ be a hyperplane arrangement in $\mathbb{R}^n$ with rank r. Then:

- The number of regions of $\mathcal{A}$ is $(-1)^n \chi_{\mathcal{A}}(-1)$;
- The number of relatively bounded regions of $\mathcal{A}$ is $(-1)^r \chi_{\mathcal{A}}(-1)$.

Corollary 3.5

The arrangements $Shi(G)$ and $Ish(G)$ have the same number of regions and the same number of relatively bounded regions.

The normals of either $Shi(G)$ and $Ish(G)$ spans the hyperplane $x_1+\cdots+x_n=0$

$Shi(G)$ and $Ish(G)$ 超平面方程形如 $x_i-x_j=c$ → normal vector is e_i-e_j

e.g. $x_2-x_5=1$ $\quad e_2-e_5=(0,1,0,0,-1,0,\cdots,0)$ $\quad V=\{(a_1,\cdots,a_n)\in\mathbb{R}^n : a_1+\cdots+a_n=0\}$.

$x_1\ x_2\ \cdots\ x_n=0$.

Hence each of these arrangements has rank $n-1$ → don't have bounded regions

their relatively bounded regions exactly have one degree of freedom

---

4. Labeling the regions

We devise combinatorial labels for the region of the deleted arrangements $Shi(G)$ and $Ish(G)$, we call Shi ceiling diagrams and Ish ceiling diagrams.

4.1 Shi ceiling diagrams

$Cox(n)$'s region is one-to-one to the symmetric group S_n

$C\subseteq\mathbb{R}^n$ is the dominant cone $(x_1>x_2>\cdots>x_n)$

The all regions of $Cox(n)$ is the set $\{wC : w\in S_n\}$, $wC \to x_{w(1)}>\cdots>x_{w(n)}$

Suppose R is a region of deleted Shi arrangement $Shi(G)$.

$Cox(n)\subseteq Shi(G)$ → $R\subseteq wC$

What's R's ceiling?

The hyperplanes of $Shi(G)$ that intersect wC are precisely

$$\Phi^+(G,w) := \{x_{w(i)}-x_{w(j)}=1 : i<j\in G \text{ and } w(i)<w(j)\}$$ → noninverse.

Suppose that $v\in R$ satisfies $v_{w(i)}-v_{w(j)}<1$ (region R is "below" some hyperplane $x_{w(i)}-x_{w(j)}=1$)

R is also below any hyperplane of the form $x_{w(i')}-x_{w(j')}=1$ s.t.

$w(i)\le w(i')<w(j')\le w(j)$ because $x_{w(1)}>x_{w(2)}>\cdots>x_{w(n)}$

We can define a partial order on $\Phi^+(G,w)$ by saying:

$x_{w(i')}-x_{w(j')}=1$ is "less" than $x_{w(i)}-x_{w(j)}=1$

when $w(i)<w(i')<w(j')<w(j)$

Then the collection of $Shi(G)$-hyperplane above R forms a down-closed set. I.

P is a poset. $a,b,c,\cdots\in P$

$b\le a$

$a\in I,\ b\le a \Rightarrow b\in I$.

Thm 4.1

There is a bijection between regions of $Shi(G)$ in the cone wC and order ideal (down-closed set) in the poset $\Phi^+(G,w)$

This map sends a region R to the set of hyperplanes in $Shi(G)$ that ar "above" R. (above : R and the origin in the same half space).

The maximal elements of the ideal are the ceiling of R.

Pf: region $\longrightarrow$ order ideal.

$\longleftarrow$ (Athanasiadis and Linusson)

R's ceilings correspond to the element of the order ideal that can remove singally. → still be an order ideal.

→ the maximal elements of ideal

To express this combinatorially,

the order ideal in $\Phi^+(G,w)$ are equivalent to Nonnesting G-partition whose blocks are "increasing" with respect to w. ~~$(w(i_1) < w(i_2) < \cdots < w(i_r)$ for $i_1 < i_2 < \cdots < i_r)$~~ $(i<j \quad w(i)<w(j))$

order ideal $\longleftrightarrow$ its maximal elements $\longleftrightarrow$ a nonnesting G-partition.

(antichain) (every $i<j$ s.t. $w(i)<w(j)$).

$(x_{w(i)} - x_{w(j)} = 1 \longleftrightarrow$ arc $ij)$.

$R \longleftrightarrow I_R \longleftrightarrow Max(I_R) \longleftrightarrow$ partition arc diagram

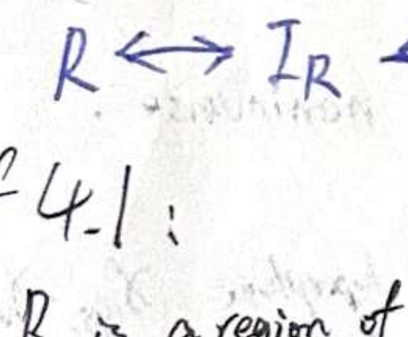
Def 4.1:

R is a region of $Shi(G)$, $R \subseteq wC$, we associate R with the pair (w, π) (order ideal ↓ a nonnesting G-partition whose blocks are increasing with respect to w.)

We draw (w,π) by placing the arc diagram for π above the number $w(1), \cdots, w(n)$.

↳ Shi ceiling diagram of R.

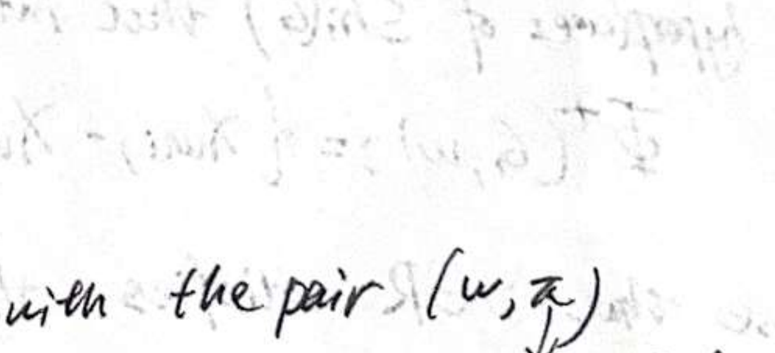
eg. $K_8 = \binom{[8]}{2}$, $w = 5\,1\,2\,8\,6\,3\,4\,7$ (1 2 3 4 5 6 7 8)



If $i<j$, $w(i)<w(j)$ → □

maximal elements → ◎

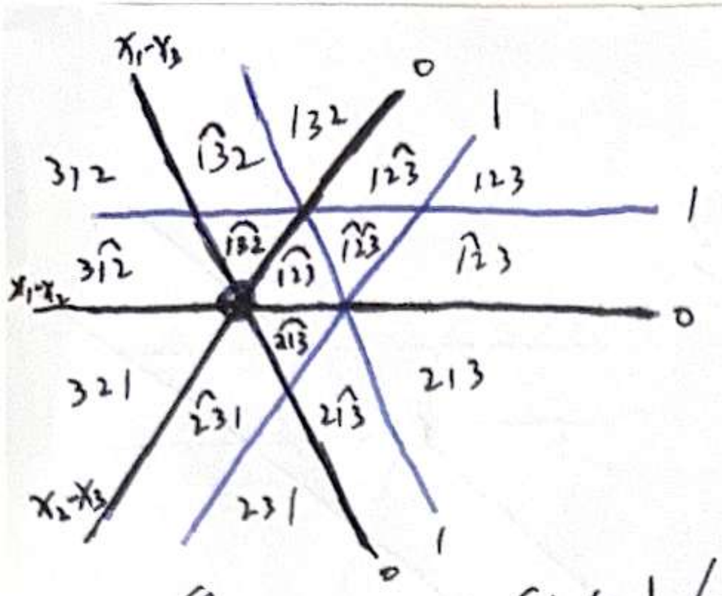
the elements in the order ideal → ◉



Shi ceiling diagram.

$x_5 - x_8 = 1 \quad x_1 - x_6 = 1 \quad x_3 - x_7 = 1$

$\pi = \{\{1,4\},\{2,5\},\{3\},\{6,8\},\{7\}\}$



The Shi arrangement Shi(3) labeled by ceiling diagrams.

Lem 4.2.

Let R be a region of $Shi(G)$ with ceiling diagram (w,π).

This region has ~~of~~ d degree of freedom iff the nonnesting partition π of $[n]$ has d connected components.

Pf: Suppose π has d connected components.

i.e. $\exists\ 1<i_1<\cdots<i_{d-1}<n$ s.t. π refines the partition

$$\{\{1,2,\cdots,i_1-1\},\{i_1,\cdots,i_2-1\},\cdots,\{i_{d-1},\cdots,n\}\}$$

Then compute $Rec(R)$:

$v=(v_1,\cdots,v_n)\in Rec(R)$

Since $R\subseteq wC \Rightarrow v_{w(1)}\geq v_{w(2)}\geq\cdots\geq v_{w(n)}$.

If $ij\in G$ and ~~connected component~~ block i,j in the same ~~block~~ block, then i,j is arc of π.

$x_{w(i)}-x_{w(j)}<1$ holds on R

$x_{w(i)}-x_{w(j)}=1$ is ceiling of R.

$$(x_{w(i)}+t v_{w(i)})-(x_{w(j)}+t v_{w(j)})=(x_{w(i)}-x_{w(j)})+t(v_{w(i)}-v_{w(j)})<1$$

$$\Rightarrow v_{w(i)}\leq v_{w(j)} \Rightarrow v_{w(i)}=v_{w(j)}$$

$v_{w(i)}=\cdots=v_{w(j)}$

$\Rightarrow Rec(R):=\{w(a_1,a_2,\cdots,a_n): a_1\geq\cdots\geq a_n\}$

If i,j in the same connected component, the $a_i=a_j$

$\Rightarrow \dim(Rec(R))=d$. □

e.g. $w=51286347$ region R.

5 | 1 2 8 6 | 3 4 7

1 2 3 4 5 | 6 7 8

The connected components of π $\{1,2,3,4,5\}$ $\{6,7,8\}$

$w(\pi)$ $\{51286\}$ $\{347\}$

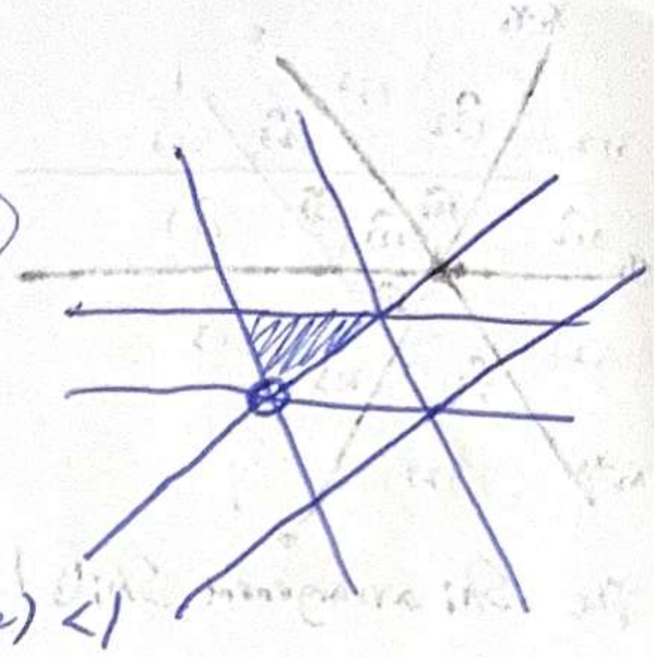
a a b b a a b a

1 2 3 4 5 6 7 8

e.g. $w=132$ $wC=\{x_1>x_3>x_2\}$. $n=3$.

take $R=\{x_1>x_3>x_2,\ x_1-x_2<1\}$ ($x_1-x_3<1$) $\longrightarrow$ $x_{w(1)}-x_{w(3)}<1 \Rightarrow (13)$

$\pi=\{\{13\},\{2\}\}$



$\Rightarrow \widehat{132}$ $d=1$

recession cone: Since $x_1>x_3>x_2 \Rightarrow *v_1\geq v_3\geq v_2$

$x_1-x_2<1$ $\quad (x_1+tv_1)-(x_2+tv_2)<1$ $\quad (x_1-x_2)+t(*v_1-*v_2)<1$

$\Rightarrow v_1-v_2\leq 0$ $\Rightarrow v_1=v_2$ $\Rightarrow v_1=v_2=v_3$

$Rec(R)=\{(a,a,a): a\in\mathbb{R}\}$ $\dim(Rec(R))=1$

$\Rightarrow d=\dim(Rec(R))$.

$n=4$. $w=2134$ $wC=\{x_2>x_1>x_3>x_4\}$

$R=\{x_2>x_1>x_3>x_4,\ x_1-x_3<1,\ x_3-x_4<1,\ x_1-x_4>1,\ x_2-x_1>1,\ x_2-x_3>1,\ x_2-x_4>1\}$

$x_{w(2)}-x_{w(3)}<1 \rightarrow 23$

$x_{w(3)}-x_{w(4)}<1 \rightarrow 34$

$\pi=\{\{1\},\{234\}\}$.

~~1234~~ $2\widehat{1}\widehat{34}$ $d=2$.

$x_2\geq x_1\geq x_3\geq x_4$ $\quad x_1+tv_1-x_3-tv_3<1 \Rightarrow v_1\leq v_3 \Rightarrow v_3=v_1$

$Rec(R)=\{(a,b,b,b): a,b\in\mathbb{R}\}$ $\dim(Rec(R))=2$.

4.2 Ish ceiling arrangement.

$Cox(n) \subseteq Ish(G)$ every region R of $Ish(G)$ is contained in wC, $\forall v=(v_1,\cdots,v_n) \in R$, $v_{w(1)} > \cdots > v_{w(n)}$

If $x_i - x_j = i$ intersects the cone wC, $\Rightarrow$ ~~$x_i > x_j$~~ on wC.

$\Rightarrow w^{-1}(i) < w^{-1}(j)$

$\Rightarrow \Psi^+(G,w) := \{ x_i - x_j = i : ij \in G,\ w^{-1}(i) < w^{-1}(j) \}$

$i<j$. $w^{-1}(i) < w^{-1}(j)$

Let R be a region of $Ish(G)$ in wC and suppose R is "below" $x_i - x_j = i$ (i.e. each $v \in R$, $v_i - v_j < i$). Then R is also below $x_{i'} - x_{j'} = i'$ where

$i \le i'$, $w^{-1}(j') \le w^{-1}(j)$ either $i < i'$ or $w^{-1}(j') < w^{-1}(j)$. (*)

at least one inequality is strict.

Define a partial order on $\Psi'(G,w)$

If (*) holds, $x_i - x_j = i$ is "less than" $x_i - x_{j'} = i'$

Thm 4.3

There is a bijection between regions of $Ish(G)$ in the cone wC and order filters (up-closed sets) in the poset $\Psi^+(G,w)$

F
$x \in F,\ x \le y \Rightarrow y \in F$

The map sends R to the set of hyperplanes in $Ish(G)$ that are "above" R (above: contain R and origin in the same half-space).

The minimal elements of the filter are the ceilings of R.

Pf: ~~I~~

$R \subseteq wC$

The collection of $Ish(G)$-hyperplanes above R is and order filter in $\Psi^+(G,w)$

~~$R \subseteq wC$ can determine~~

The collection can determine R. → ~~injection~~. injective.

R's ceilings is are precisely the elements of this filter that may be individually removed to obtain another filter. → minimal elements.

To show surjective. filter ⇒ region

Let $F \subseteq \Psi^+(G,w)$ be an order filter. A is its set of minimal elements.

For $1 \le i \le n$ .def: $z_i := -\max\{ j : x_1 - x_k = j \in A \text{ and } w^{-1}(k) \le w^{-1}(i) \}$.

$(z_{w(1)}, \cdots, z_{w(n)}) \in \mathbb{R}^n$ lies on the boundary of a region of $Ish(G)$, which maps to F

take the minimal $x_1 - x_k = j \in A$

consider the minimal ~~*~~ at the left of $x_1 - x_k = j$: $x_1 - x_{k'} = j'$ s.t. $j' < j$ because can't compare.

$\Rightarrow \max\{ j' : x_1 - x_{k'} = j' \in A,\ w^{-1}(k') \le w^{-1}(k) \} = j \Rightarrow z_k = -j$, $z_1 = 0$ ($w^{-1}(1) < w^{-1}(j)$, $w^{-1}(k) \le w^{-1}(1)$, $\phi \Rightarrow \phi$)

$z_1 - z_k = j$

From Thm 3.2. $\chi_{Ish(G)}(p) = p\sum_{k=0}^{n-1}(-1)^k Stir(G,n-k)\frac{(p-k-1)!}{(p-n)!}$

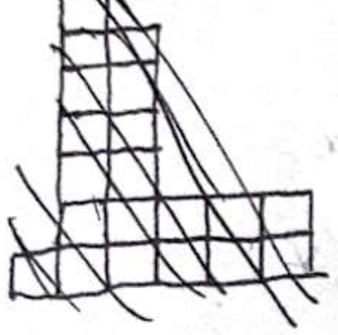
Zaslavsky Thm: $\#\{\text{region}\} = (-1)^n\chi(-1) = \sum_{k=0}^{n-1} Stir(G,n-k)\frac{n!}{(k+1)!}$

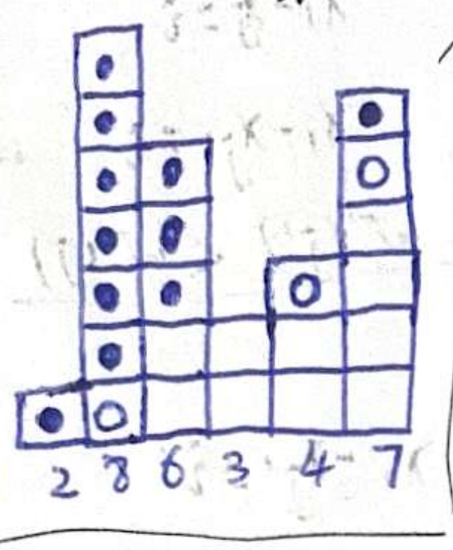
Thm 5.1: Fixed a ceiling partition (a,b) which has r blocks, the number of regions is $\frac{n!}{(n-r+1)!}$

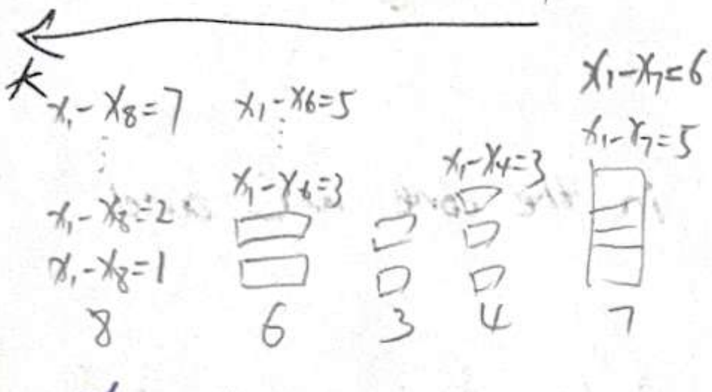
let $r=n-k$, $n-k$ blocks, k arcs. G-partition's number is $Stir(G,\underline{n-k})$ 块

For fixed k, totally $Stir(G,n-k)\frac{n!}{(k+1)!}$

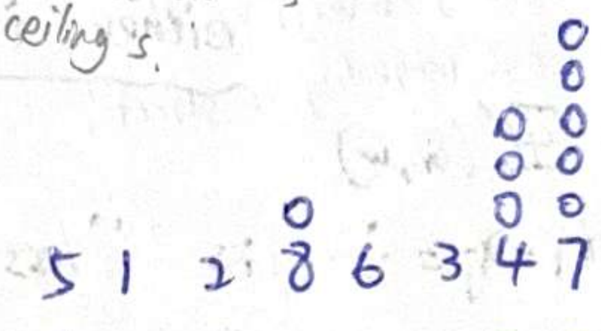
$\sum_{k=0}^{n-1} Stir(G,n-k)\frac{n!}{(k+1)!}$ $= \#\{\text{filters summed over } w\}$.







order filter
○ minimal elements
ceiling's.



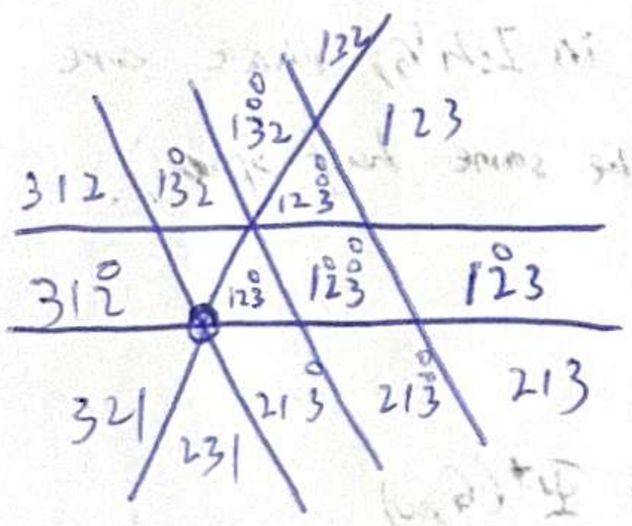
Ish ceiling diagram

ε: (0 0 0 1 0 0 3 5)

$(w,\varepsilon) = (5128634\,7, (00010035))$

Ish ceiling diagram

- $0 \le \varepsilon_i < w(i)$.
- $\varepsilon_i = 0$ unless $w^{-1}(i) < w^{-1}(i)$
- if $\varepsilon_i > 0$, then $\varepsilon_i < w(i)$ is an edge in G.
- the nonzero entries of ε strictly increase.



$x_i - x_{w(i)} = \varepsilon_i$

Lem 4.4.

Let R be a region of $Ish(G)$ in the cone wC with ceiling diagram $(\sharp w, \varepsilon)$

If k is the maximum index s.t. $\varepsilon_k \neq 0$ (or $k = w^{-1}(1)$ if ε is $\vec{0}$),

then R has $n - k + w^{-1}(1)$ degrees of freedom.

R is relatively bounded iff $w(1) = 1$ and $\varepsilon_n \neq 0$.
↳ $k = n$

Pf: $v = (v_1, \cdots, v_n) \in Rec(R)$, $R \subseteq wC$

$\Rightarrow v_{w(1)} \geq v_{w(2)} \geq \cdots \geq v_{w(n)}$ R位于: $x_1 - x_{w(j)} < \varepsilon_j$

If $\varepsilon_j \neq 0$, must have $v_1 - v_{w(j)} \leq 0$ $\Rightarrow v_{w(w^{-1}(1))} \leq v_{w(j)}$

$w^{-1}(1) < j \Rightarrow v_{w(w^{-1}(1))} \geq \cdots \geq v_{w(j)}$ 角标是在w中的位置，不是坐标编号 ↳ $v_1 = v_{w(w^{-1}(1))} \geq v_{w(w^{-1}(1)+1)}$

$\Rightarrow v_{w^{-1}(1)} = v_{w^{-1}(1)+1} = \cdots = v_{w^{-1}(j)}$ $v_{w(w^{-1}(1))} = v_{w(w^{-1}(1)+1)} = \cdots = v_{w(j)}$

$\Rightarrow Rec(R) = \{ w(a_1, a_2, \cdots, a_n) : a_1 \geq \cdots \geq a_n,\ a_{w^{-1}(1)} = a_{w^{-1}(1)+1} = \cdots = a_k \}$.

$\dim(Rec(R)) = n - (k - w^{-1}(1) + 1) + 1 = n - k + w^{-1}(1)$ ■

$w = 5\,1\,2\,8\,6\,3\,4\,7$

$w(1) = 5$ $k = 8$. $\varepsilon_k \neq 0$.

$v_5 \geq v_1 \geq v_2 \geq v_8 \geq v_6 \geq v_3 \geq v_4 \geq v_7$

$w^{-1}(1) = 2$ $\varepsilon_8 = 5$

$v_{w(2)} = v_{w(3)} = \cdots = v_{w(8)}$

$v_1 = v_2 = v_8 = v_6 = v_3 = v_4 = v_7 = a$, $v_5 = b$. $(a\,a\,a\,a\,b\,a\,a\,a)$

$\dim = 2$.

4.3 Bijection between dominant regions.

Shi ceiling ~~diag~~ diagrams and Ish ceiling diagrams give us a bijection between dominant regions of $Shi(G)$ and $Ish(G)$ with same number of regions.

But. bijection not preserve degree of freedom.

Thm 4.5

$G \subseteq \binom{[n]}{2}$, integer c.

The deleted arrangements $Shi(G)$ and $Ish(G)$ have the same number of dominant regions with c ceilings.

Pf: $w=1$. $\mathcal{D}^+(G,1) = \{x_i - x_j = 1 : ij \in G\}$

$\mathcal{I}^+(G,1) = \{x_1 - x_j = i : ij \in G\}$. ■

Note that the bijection doesn't extend to other cones wC.